Electric Motor-Driven Hydraulic Pump – Maintenance Practices

1. Adjustment/Test

Equipment required:

- Suitable Test Rig with components as shown in Figure 1 filled with clean fluid (MIL–H–5606 or MIL–H–6083, Type 1)

A. Bench Test Electric Motor-Driven Hydraulic Pump

Refer to Figure 1 for details.

1) Remove hydraulic power package. (Refer to 29-10-00, Hydraulic Power Package (Mod 6/2039) – General Data).

2) Remove hydraulic pump with motor from hydraulic power package. (Refer to Para 2.A).

3) Check for obvious damage.

4) Check electric motor brush wear (3/16 inch minimum allowable remaining brush length).

5) Connect pump into circuit of test rig. Secure to test bench.

6) Connect 18 to 28 Volt DC power supply and switch power on.

7) Bleed all air from pump.

8) Close valve B slowly until 1500 psi pressure is obtained on the discharge pressure indicator. Regulate voltage supply to 27 ± 0.5 volts.

9) Run pump for 6 minutes. Check the following:

a) Leakage from seal drain port does not exceed 0.5 cc.

b) Current draw does not exceed 33 amperes.

c) Discharge flow is 0.42 Imp (0.50 U.S.) gpm minimum.

10) Open valve B until pressure reads 1000 psi.

11) Slowly close valve A until suction line depression reads – 1.5 inches Hg. Regulate supply voltage to 18 + 0.5 or – 0 volts. Switch off electrical supply.

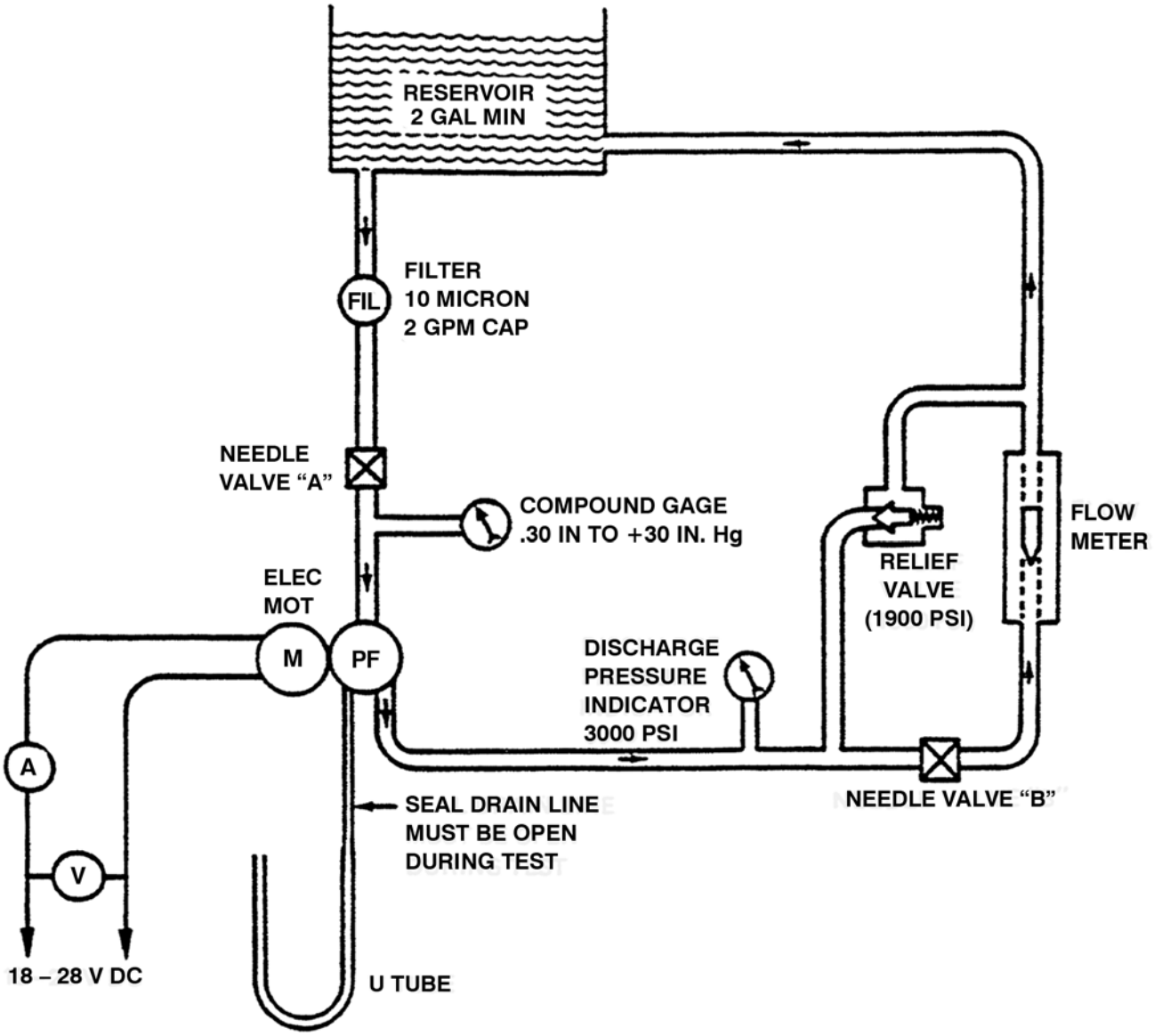
12) Allow pump to stand for 2 minutes.

13) Without altering any settings, switch on electrical supply. Check that pump discharge pressure rapidly builds up to 1000 psi.

14) Switch off electrical supply. Open valve B and release pressure.

15) Remove pump from test rig.

16) If required for service, reinstall pump on hydraulic power package. If to be placed in storage, fill pump with inhibiting fluid (MIL–H–6083 Type 1), and fit dust plugs.



Suggested Testing Rig
Figure 1

2. Removal/Installation

A. Removal of Hydraulic Pump and Motor

Refer to 29-00-00, Hydraulic Power – General Data.

1) Remove both inlet and outlet hydraulic lines from pump.

2) Remove the four AN4 bolts.

3) Lift pump motor clear of power package.

4) Clean any excess hydraulic fluid off the power package pallet.

5) Disconnect the Pressure Switch on the power package.

6) Disconnect the MS power connector from the noise filter of motor.

7) Disconnect the Thermal Switch connector.

B. Installation of Hydraulic Pump and Motor

Instruction to install the hydraulic pump and motor and connection of Thermal Switch:

1) Set the hydraulic pump and motor in place on the power package pallet.

2) Install the four AN4 bolts, washer and nuts, finger tight.

3) Connect both inlet and outlet lines to pump.

4) Torque the four AN4 bolts to 35 – 40 in–lbs.

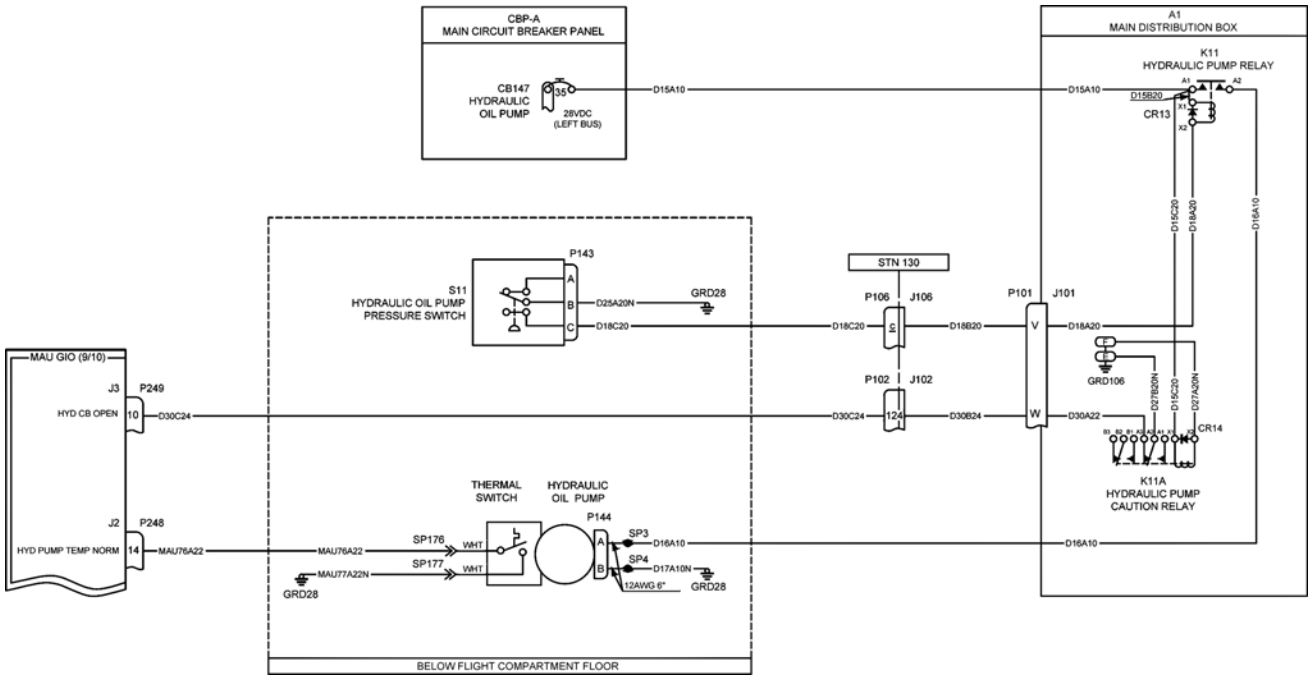
5) Connect Pressure Switch on the power package.

6) Connect the MS power connector to the noise filter of motor.

7) Connect the Thermal Switch connector.

8) Refer to Figure 2 for wiring reference.

The Thermal Switch is mounted to the motor near the shaft bearing area and senses the temperature of the motor cap end. The Thermal Switch is a normally closed switch. When the temperature reaches 180°F ± 5°F, the switch will open and provide an indication in CAS. When the temperature reaches 165°F ± 5°F, the switch will close and the indication will be extinguished.



Hydraulic System Electrical
Figure 2

C. Troubleshooting

1) Do a function test according to 29-10-00, Hydraulic Power Package (Mod 6/2039) – General Data.

2) If the motor fails to run during the test:

a) Remove the motor from the power package and test per Para 1.A.

b) If the motor runs on the above test then the motor is functioning properly.

c) If the motor fails to run on above test, remove and replace.

3) If the motor runs and the pump does not produce the required pressure:

a) Remove the pump and motor assembly and test per Para 1.A.

b) If the pump produces the required pressure and flow, the pump is working properly.

c) If the pump does not produce the required pressure and flow, remove and replace.

4) The thermal switch is tested by reading the resistance across terminals SP176 and SP177 of the two MS connectors. If the reading is more than 1.0 ohms remove and replace.